1. Without using a calculator, solve the inequality

$$\frac{9}{2-x-x^2} \le 4.$$ **[4]**

2. Show that the equation $y=\sqrt{3}\sin 2x-2\sin^2 x$ can be written as

$$y=R\sin(2x+\alpha)+A,$$

where R, A and α are constants to be found. **[3]**

Hence state a sequence of transformations which transform the graph of $y=2\sin 2x$ to the graph of $y=\sqrt{3}\sin 2x-2\sin^2 x$. **[2]**

3. **(i)** Express $1-\mathrm{i}$ in the form $re^{\mathrm{i}\theta}$, giving the values of r and θ in exact form. **[1]**

(ii) Hence find the exact value of $1+z+z^2+\ .\ .\ .\ +z^{10}$ where $z = 1-\mathrm{i}$. **[4]**

4. Relative to an origin O, points A, B and C have position vectors $\begin{pmatrix}7\\6\\-1\end{pmatrix}$, $\begin{pmatrix}3\\4\\5\end{pmatrix}$ and $\begin{pmatrix}4\\1\\7\end{pmatrix}$ respectively. The point P is such that $\overrightarrow{AB}=2\overrightarrow{BP}$. Find

(i) the position vector of P, **[2]**

(ii) angle APC, **[3]**

(iii) the exact value of the area of triangle APC. **[2]**

5. A sequence $u_1,\ u_2,\ u_3,\ ....$ is such that $u_1=\frac{1}{2}$ and

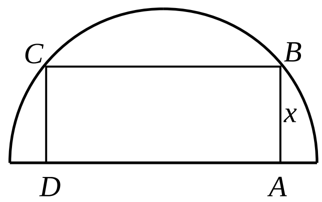
$$2u_{n+1}=u_n+2^{1-n},\ \text{for all } n\ge 1.$$

(i) Find the values of u_2, u_3 and u_4. **[2]**

(ii) Give a conjecture for u_n in terms of n. **[1]**

(iii) Prove your conjecture by induction. **[4]**

6.



The diagram shows a rectangle $ABCD$ inscribed in a semi-circle with fixed radius r cm. Two vertices of the rectangle lie on the arc of the semi-circle. If $AB = x$ cm, show that the perimeter P of the rectangle $ABCD$ is $2x + 4\sqrt{r^2 - x^2}$. **[2]**

Given that as x varies, the maximum value of P occurs when $AB : BC = 1 : k$, find k. **[4]**

7. **(i)** Express $\dfrac{1}{r(r+2)}$ in partial fractions. **[2]**

(ii) Hence or otherwise, find, in terms of n, $\sum_{r=1}^{n} \dfrac{1}{r(r+2)}$. **[2]**

(iii) State the sum to infinity. **[1]**

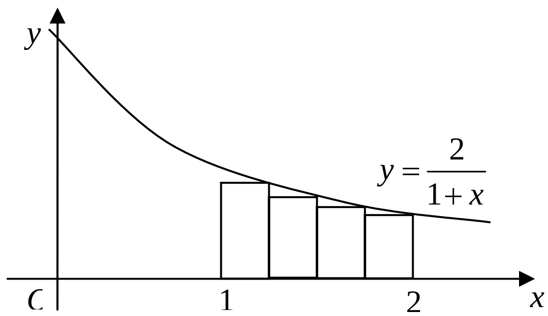
(iv) Deduce the exact sum of the series

$$\frac{1}{48} + \frac{1}{63} + \frac{1}{80} + \ldots.. + \frac{1}{399}.$$ **[2]**

8. The parametric equations of a curve are $x = \sin^3\theta$, $y = \cos^3\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Find the equation of the tangent at the point $P\left(\sin^3 p, \cos^3 p\right)$. **[3]**

The tangent at the point P meets the x-axis at T and the y-axis at N. State the coordinates of T and N as single trigonometric functions in terms of p, and hence show that the area of triangle $OTN = \frac{1}{4}\sin(2p)$. **[5]**

9. The diagram shows the parts of the graph of $y = \dfrac{2}{1+x}$ between $x = 1$ and $x = 2$. Prove that the total area of the four rectangles, each of equal width, as shown in the diagram below, may be expressed as $\displaystyle\sum_{r=1}^{4} \frac{2}{8+r}$. **[2]**



In general, when there are n rectangles, each of equal width, under the curve between $x = 1$ and $x = 2$ (instead of just 4), find an expression, in the form $\displaystyle\sum_{r=1}^{n} \mathrm{f}(r)$, for their total area.

[2]

Hence show that $\displaystyle\sum_{r=1}^{n} \frac{2}{2n+r} < 2\ln\left(\frac{3}{2}\right)$. **[2]**

Deduce that $\displaystyle\sum_{r=1}^{n} \frac{2}{2n+r} < 2\ln\left(\frac{3}{2}\right) < \sum_{r=1}^{n} \frac{2}{2n+r-1}$. **[2]**

10. (a) Use the fact that $5\sin x - 3\cos x = (\cos x + \sin x) - 4(\cos x - \sin x)$ to find the exact value of $\displaystyle\int_{\frac{\pi}{6}}^{0} \frac{5\sin x - 3\cos x}{\cos x - \sin x}\,\mathrm{d}x$. **[4]**

(b) Find $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(x\sqrt{1-x^2}\right)$, expressing your result in the form $\dfrac{a+bx^2}{\sqrt{1-x^2}}$, where a and b are constants to be determined. **[2]**

Hence, or otherwise, obtain the exact value of $\displaystyle\int_{0}^{\frac{1}{2}} \frac{3-2x^2}{\sqrt{1-x^2}}\,\mathrm{d}x$. **[3]**

11. Three functions f, g, h are defined as follows :

$$\mathrm{f}: x \mapsto 2^x, \qquad x \in \mathbb{R}$$
$$\mathrm{g}: x \mapsto 2^{x-1}, \qquad x \in \mathbb{R}$$
$$\mathrm{h}: x \mapsto x^2 + x - 2, \qquad x \in \mathbb{R}$$

(i) Describe a single geometrical transformation which maps the graph of f onto the graph of g. **[1]**

(ii) Sketch the graph of h and explain why h^{-1} does not exist. **[2]**

(iii) Write down the value of a such that $\{x \in \mathbb{R}, x > a\}$ is the largest possible domain of h for which h^{-1} exists, and hence define h^{-1} in a similar form. **[4]**

(iv) Using the domain of h found in (iii), explain why $\mathrm{h}^{-1}\mathrm{f}$ exists. Define $\mathrm{h}^{-1}\mathrm{f}$ in a similar form and find its range. **[4]**

12. A curve C has the equation $y = \dfrac{ax^2 + bx + c}{x + d}$ for some constants a, b, c and d. Given that the asymptotes are $x = 5$ and $y = x + 7$, find the values of a, b and d. **[3]**

Given that the curve has a stationary point at $x = -1$, find the value of c. **[2]**

Draw a sketch of C, indicating clearly the asymptotes, the coordinate(s) of the stationary points and the point(s) of intersection with the axes. **[3]**

By drawing a sketch of another suitable curve on the same diagram, state, with a reason, the number of roots of the equation

$$(1-\lambda)x^2 + (2+5\lambda)x + 1 = 0$$

(i) when $\lambda = 1$,

(ii) when $0 < \lambda < 1$. **[3]**

13. (a) A sample of radium loses mass, by disintegration, at a rate which is proportional to its mass x at that instant. Write down a differential equation relating x and t, where t is the number of years from the start of the disintegration.

Given that it would take 1600 years for the sample to lose half of its initial mass, find how long (to the nearest year) it takes to lose 1% of the initial mass. **[6]**

(b) Solve the differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = \frac{1}{1+4t^2} \quad ,$$

given that when $t = 0$, $x = 0.5$ and $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 0$. **[6]**

End of Paper